Source model of the 2026 M_w 6.9 North Sumatra intraslab earthquake

A 172-km-deep rupture beneath the Toba region — and the cleanest test yet of the teleseismic MT+FFI pipeline

Automated pipeline (17.Venezuela_2026)

2026-08-15 — no USGS finite-fault product exists for this event

Summary

The M_w 6.9 earthquake of 2026-08-15 beneath north Sumatra ruptured the subducted Indo-Australian (Sunda) slab at 172–181 km depth, below the Toba caldera region — an intraslab event, not a megathrust rupture. With 231 teleseismic stations at a 21° maximum azimuthal gap and a 98.6 % double-couple source, this is the pipeline's cleanest case: body-wave misfits of 0.080 (lowest of any event studied here), M_0 recovered to <1 % of the W-phase value, and an independent MTUQ mechanism within 7.4° (Kagan) of the W-phase solution — also the best of any event here. It is the largest deep earthquake recorded within 150 km of its epicentre (previous maximum M_w 6.3, 2006).

1 Event and setting

Origin	2026-08-15T10:54:51.8 UTC, 3.086°N, 99.015°E, depth 172.5 km
Magnitude	M_w 6.9 (W-phase); $M_0 = 2.62 \times 10^{19}$ N m; centroid 180.5 km; DC 98.6 %
W-phase NP1	207.96/58.15/1.48 (left-lateral, moderately dipping)
W-phase NP2	117.18/88.75/148.14 (near-vertical, right-lateral oblique)
W-phase centroid	3.080°N, 99.013°E, origin +10.9 s (0.6 km offset)
Setting	intraslab, Sunda slab beneath the Toba caldera region

The hypocentre–centroid check (Colombia lesson) is benign here: a 0.6 km spatial offset with a +10.9 s time shift is a normal centroid lag for an M_w 6.9 with a ∼7 s half duration, not a delayed main rupture, so the origin-anchored inversion is valid.

Regional significance. ComCat 1980–2026 within 0.5–5.5°N, 96.5–101.5°E lists 718 events $M_\mathrm{w}\geq4$ deeper than 60 km (266 deeper than 150 km): a continuous, active intermediate-depth zone. Within 150 km of this epicentre and deeper than 120 km the largest prior event was M_w 6.3 (2006-12-01, 204 km depth, 35 km away), with all others $M_\mathrm{w}\leq5.6$ — this event releases roughly eight times that moment.

2 Data

231 three-component stations, 31–90°, maximum azimuthal gap 21° — the best coverage of any event in this project. The mechanism is a strike-slip pair, and clustered azimuths are known to alias four-lobed strike-slip radiation (the 2026 Kumamoto lesson); a 21° gap removes that risk. At 180 km the fault lies far below WISP's ∼126.5 km surface-wave Green's-function cap, so surface waves are dropped on both planes and the two-plane comparison is body-only and fair by construction.

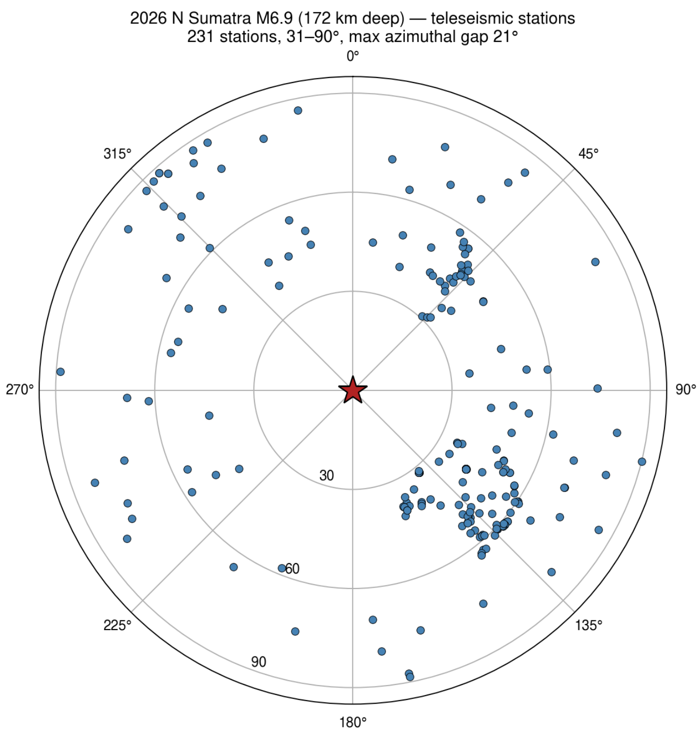


Figure 1: Teleseismic stations (azimuthal-equidistant; rings 30/60/90°).

3 Moment tensor (MTUQ)

variant	mechanism	M_w	Kagan vs W-phase
DC body P+SH @180.5 km	202/56/−7	6.75	7.4°
DC joint @180.5 km	202/49/2	6.75	11.3°
depth-scan solution at 180.5 km	207/58/−4	6.75	5.5°

Table 1: The best independent mechanism recovery in this project. For contrast, the 2026 Colombia M_w 7.4 gave 52–95° (point source broken by source complexity) and the 2023 Mariana M_w 6.1 gave 52–86° (signal-to-noise floor). Here a simple 98.6 %-DC source, a 21° azimuthal gap and the raised deep-event body band (25–60 s) combine to recover the W-phase mechanism almost exactly. M_w 6.75 versus 6.9 is a modest underestimate.

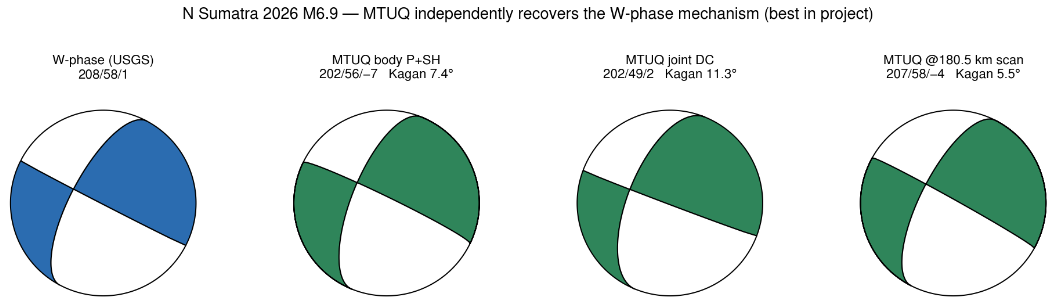


Figure 2: W-phase reference and the three MTUQ recoveries.

4 Finite fault (WISP): the fault had to be enlarged

First-pass `auto_model` fits were excellent (misfit 0.080–0.082) but violated the edge-containment rule: 29 % (NP1) and 47 % (NP2) of peak slip lay in the deepest fault row. Following the San Pedro de Atacama rule the fault was enlarged identically on both planes (69×57 km → 97×114 km) and both were re-inverted and re-shifted.

	first pass (69×57 km)	enlarged (97×114 km)
edge slip top / bottom	2–4 % / 29–47 %	2–3 % / 2–4 %
misfit NP1 / NP2	0.0816 / 0.0803	0.0829 / 0.0795
plane margin	1.6 % (meaningless)	4.1 %, NP2 preferred
peak slip	0.53–0.57 m	0.44–0.50 m
$\Delta\sigma$ (effective-area)	1.2 MPa	0.65–0.83 MPa

Table 2: The enlargement changed the conclusion: with the boundary artefact removed the plane margin grew from a meaningless tie to a suggestive 4.1 % favouring NP2 (208/58/1) — the plane MTUQ independently converged on. Both models recover $M_0 = 2.64 \times 10^{19}$ N m (<1 % from the W-phase) with $T_{5-95} \approx 15.5$ s, and the source-time function is plane-independent ($r = 0.972$): well-determined moment release on a weakly-determined plane.

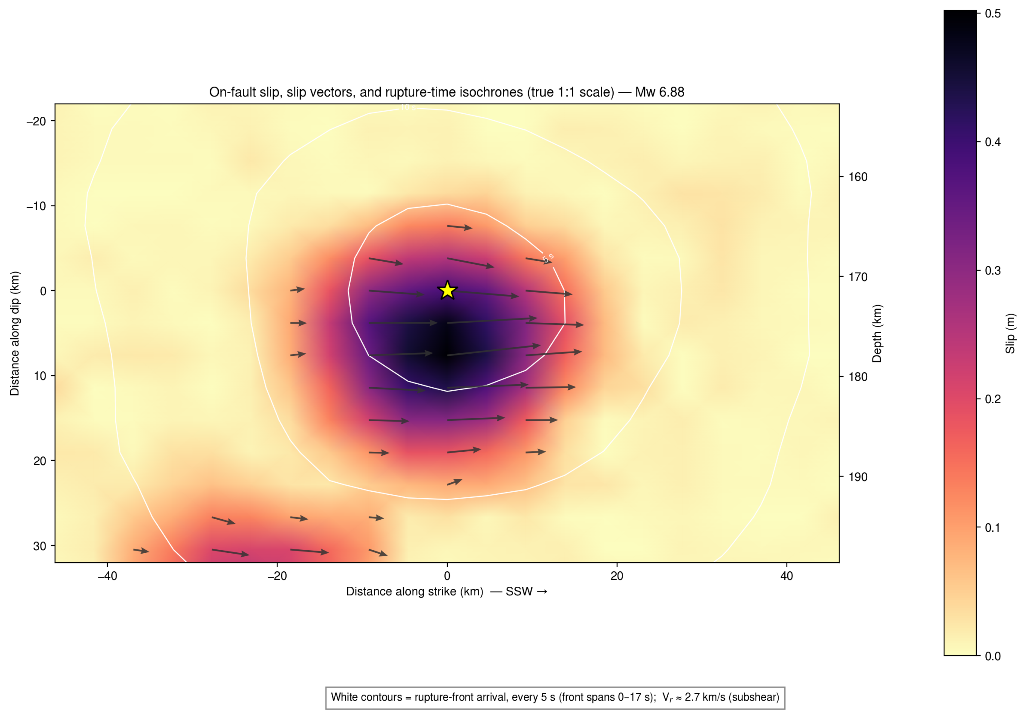


Figure 3: Preferred plane (208/58/1), true scale: compact slip at 165–207 km depth, peak 0.50 m, cleanly contained after enlargement (edge slip 2 %).

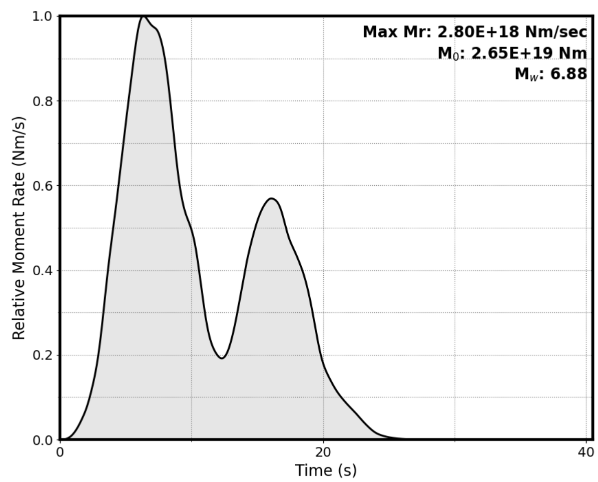


Figure 4: Moment-rate function (preferred plane).

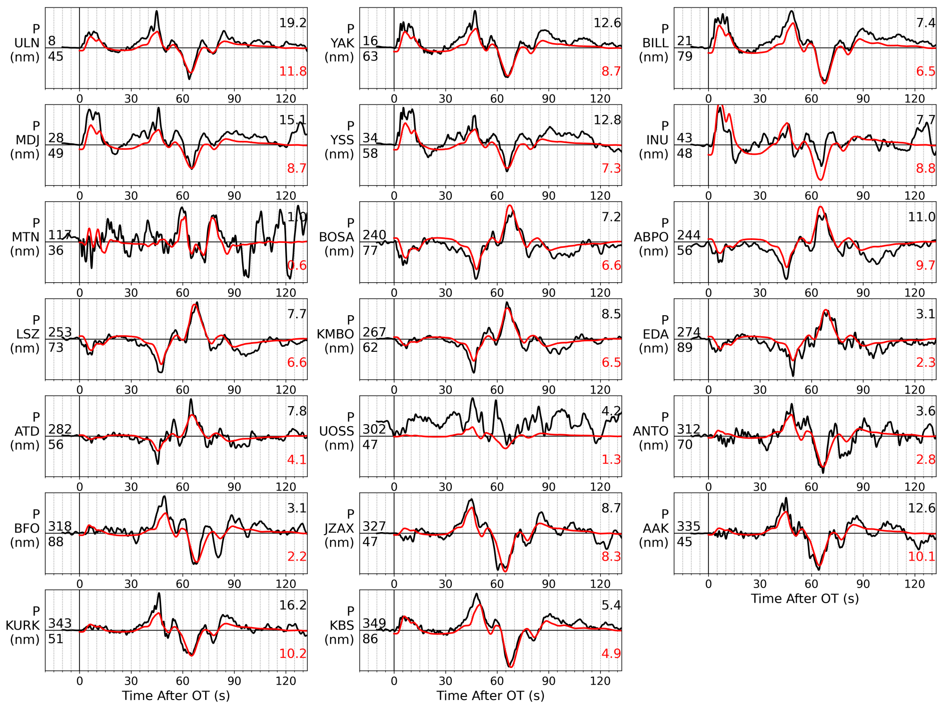


Figure 5: Teleseismic P fits (preferred plane, shifted).

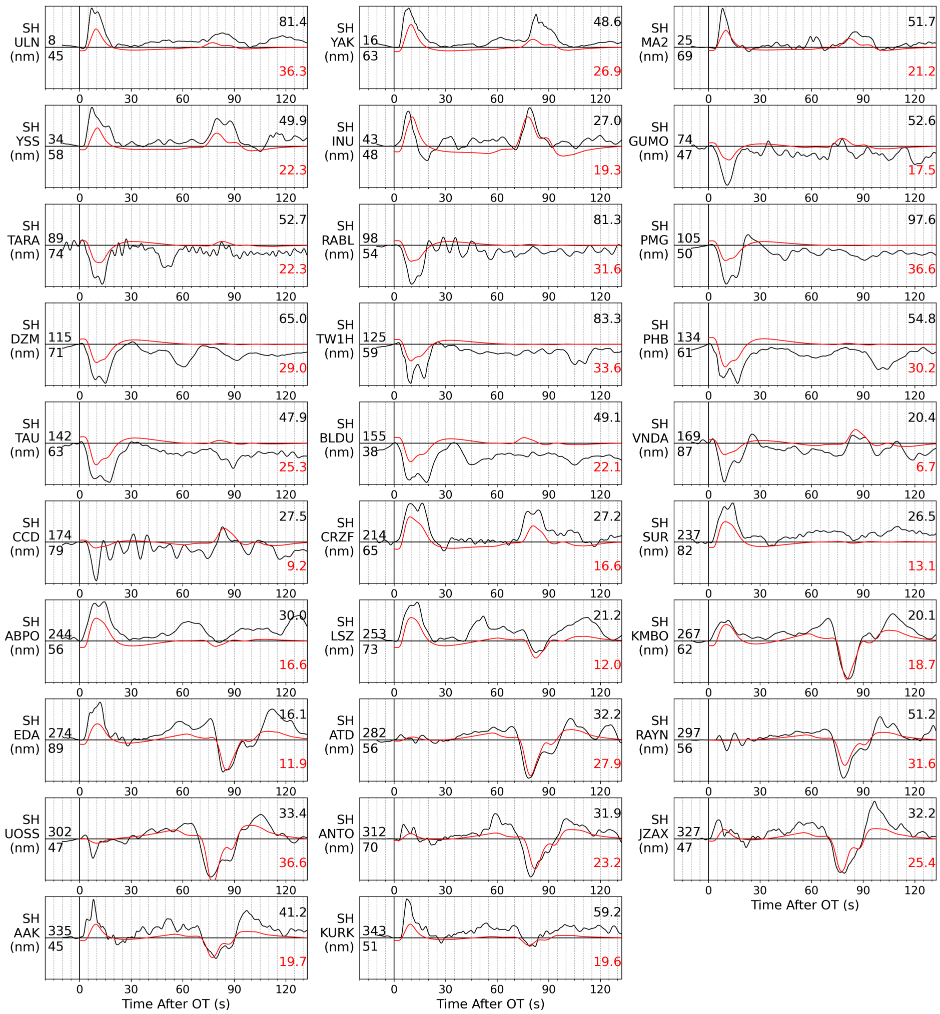


Figure 6: Teleseismic SH fits (preferred plane, shifted).

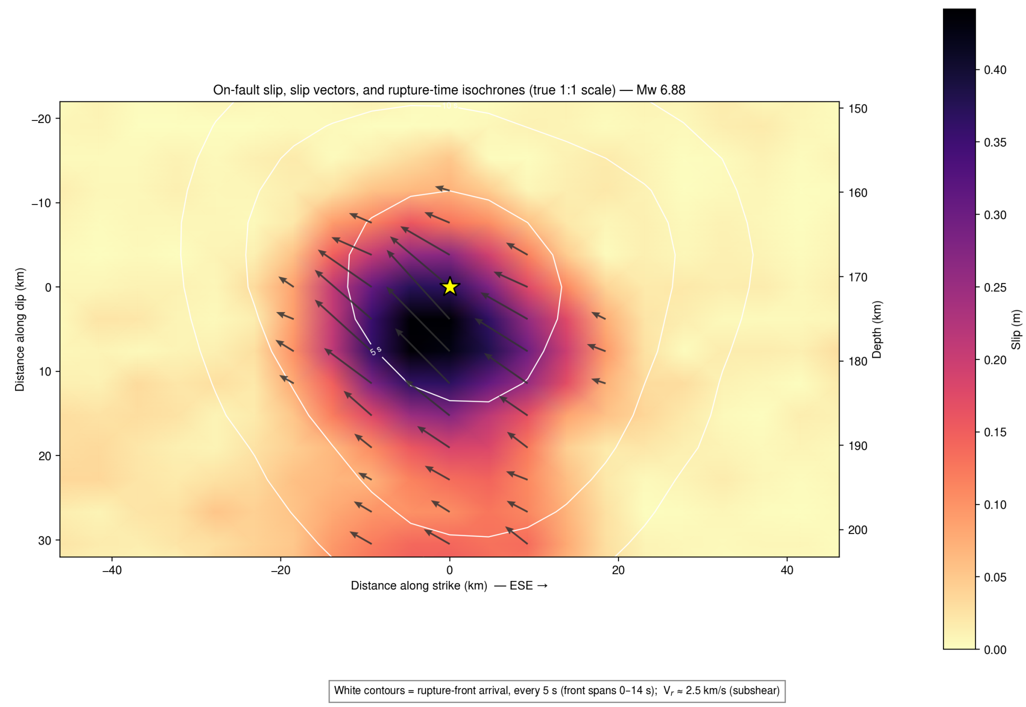


Figure 7: Conjugate plane (117/89/148) for comparison: 4.1 % worse fit.

5 Assessment

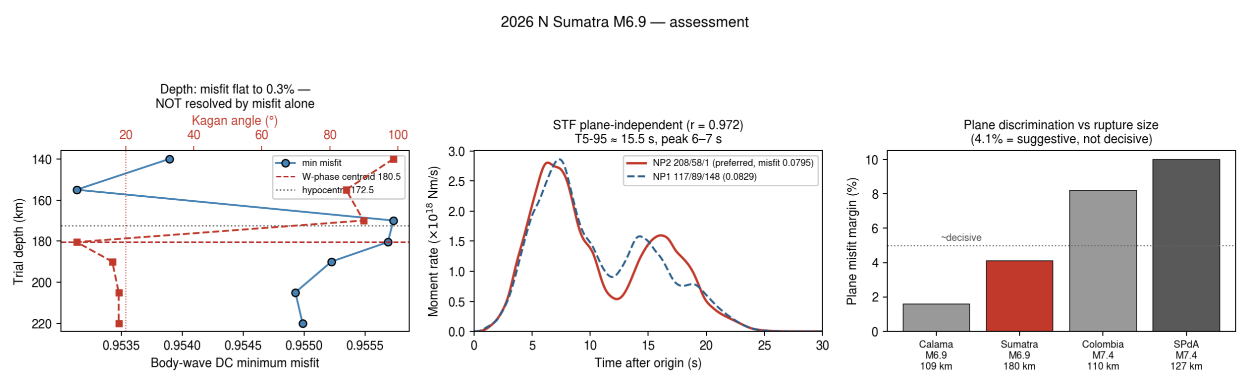


Figure 8: Left: the depth scan — misfit (blue) is flat to 0.3 % across 140–220 km, but the Kagan angle (red) reveals a sharp transition: the mechanism is chaotic at 140–170 km (85–99°) and locks onto the W-phase from 180.5 km downward (5.5–18°). Centre: the source-time function is plane-independent. Right: the plane-discrimination margin in the project's rupture-size context.

Depth — a split verdict worth stating carefully. The body-wave misfit minimum does not resolve depth (0.3 % variation over an 80-km range; its nominal minimum at 155 km is meaningless at that flatness). What does resolve it is mechanism stability: the body waves only fit coherently once the depth phases are correctly timed, which occurs at ≥180 km. Together with the WISP slip centroid at 181 km, this independently corroborates the USGS 180.5 km centroid over the 172.5

km hypocentre.

Kinematics (preferred plane). The slip centroid lies 8.4 km from the epicentre at azimuth 349°, 181 km deep; the moment centroid migrates 13.7 km toward azimuth 354° (north) while deepening from 176 to 188 km, with 66 %/23 % forward/backward moment — a moderately unilateral, northward and down-dip rupture. The realized rupture-front speed is 2.52 km s^{-1} (16–84 %: 2.24–2.95), giving $V_r/V_s \approx 0.55$ at $V_s \approx 4.6$ km s^{-1}: the same slow, dissipative regime found for the 2026 Colombia M_w 7.4 (0.56), and matching the slow first stage rather than the fast runaway stage of Poli et al. (2016). The stress drop, 0.65–0.83 MPa, is low.

6 Source-time function, asperities, stress drop, rupture speed

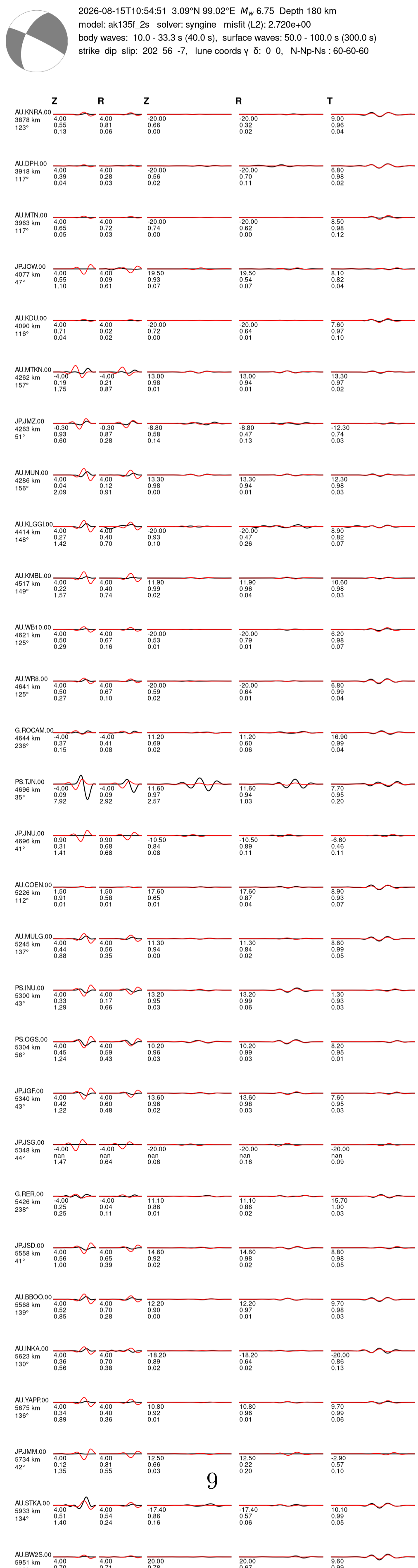
The double-peaked STF is robust; two comparable asperities are not. The second moment-rate pulse appears in four of five independently derived models (both planes × both fault sizes) and its prominence grows as the parameterization improves (0.07 and 0.16 of peak in the original edge-violating faults; 0.24 and 0.38 after enlargement). Splitting the moment by rupture time, the first pulse (t_{rup} <11 s) carries 67 % of the moment at 165–194 km depth and the second (t_{rup} ≥11 s) carries 33 % at 191–207 km; their centroids are 20 km apart (15 km horizontal, 13 km deeper) separated by 9.3 s, an apparent migration of 2.15 km s^{-1} — a legitimate rupture velocity, in contrast to the kinematically impossible 0.8 km s^{-1} of the 2026 Colombia M_w 7.4. As a spatially distinct asperity, however, the second patch is marginal: at a 50 %-of-peak threshold it is a single subfault and at 60 % it disappears. The defensible statement is a dominant asperity plus a weaker, deeper secondary pulse ∼20 km down-dip.

Stress drop depends on convention by an order of magnitude.

convention	area	radius	$\Delta\sigma$
effective, $(\Sigma u)^2/\Sigma u^2$	1827 km^2	24.1 km	0.82 MPa
slip >15 % of peak	1479 km^2	21.7 km	1.13 MPa
half-peak (>50 %)	440 km^2	11.8 km	6.98 MPa

Table 3: The spread follows from smooth slip (0.20 m mean over the effective area versus a 0.50 m peak). A slip-weighted average of the on-fault static stress-change field (Ripperger & Mai 2007 kernel, Noda et al. 2013 energy-based averaging) gives ≈0.9 MPa for the preferred model: the effective-area convention tracks the rigorous value, while the half-peak proxy overestimates it by nearly an order of magnitude. All are lower bounds, since inversion smoothing suppresses short-wavelength slip and hence stress.

Rupture speed is stable across models and interior to its allowed window (1.2–3.8 km s^{-1}, reference 3.0), so it is data-selected rather than a bound artefact: 2.52 (preferred NP2; 16–84 % 2.24–2.95), 2.60 (NP1), 2.50 and 2.38 km s^{-1} on the smaller faults. With $V_s \approx 4.6$ km s^{-1} at 180 km this gives $V_r/V_s \approx 0.55$ — the same slow, dissipative regime as the Colombia M_w 7.4 (0.56). A sensitivity test settles whether this is required or merely permitted: re-inverting the preferred model with the rupture-velocity ceiling raised from 3.75 to 4.6 km s^{-1} ($\approx V_s$) changes nothing — misfit 0.07949 in both runs (identical to five decimals), realized median 2.52 km s^{-1}, peak slip 0.50 m. Offered near-shear speeds, the data decline them: the slow rupture is required, not imposed.


2026-08-15T10:54:51 3.09°N 99.02°E M_w 6.75 Depth 180 km
model: ak135f_2s solver: syngine misfit (L2): 2.720e+00
body waves: 10.0 - 33.3 s (40.0 s), surface waves: 50.0 - 100.0 s (300.0 s)
strike dip slip: 202 56 -7, lune coords γ δ: 0 0, N-Np-Ns : 60-60-60
Z R Z R T

7 Caveats and next steps

Teleseismic-only; no USGS finite fault exists for comparison; the plane preference (4.1 %) is suggestive rather than decisive, though corroborated by the independent MTUQ mechanism; depth rests on mechanism stability rather than a misfit minimum; peak slip is smoothing-limited and therefore a lower bound. Natural next steps: a radiated-energy and efficiency measurement (does this event also sit in the low-η dissipative regime found for Colombia?); BMKG regional waveforms for an independent plane test; and comparison with the 2006 M_{w} 6.3 mechanism to test whether this slab segment ruptures repeatedly in the same geometry.